

REVTeX4

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Abstract

Contents

r=3

$$\Gamma(\hat{\lambda}_1) = \sum_{0 \leq N \leq r} \Gamma_N(\hat{\lambda}_1) = \sum_{1 \leq N \leq r} \sum_{1 \leq P \leq N} \binom{N}{P} \Gamma_P^N(\hat{\lambda}_1) = \sum_{1 \leq N \leq 3} \binom{N}{1} \Gamma_1^N(\hat{\lambda}_1) = \hat{\lambda}_1 + 2\Gamma_1^2(\hat{\lambda}_1) + 3\Gamma_1^3(\hat{\lambda}_1) \quad (1)$$

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$$\begin{aligned} & \lambda_{111} |\psi_1\rangle \langle \psi_1| + \lambda_{122} |\psi_2\rangle \langle \psi_2| + \lambda_{133} |\psi_3\rangle \langle \psi_3| + \lambda_{111} |\psi_1\psi_2\rangle \langle \psi_1\psi_2| + \lambda_{111} |\psi_1\psi_3\rangle \langle \psi_1\psi_3| \\ & + \lambda_{122} |\psi_2\psi_1\rangle \langle \psi_2\psi_1| + \lambda_{122} |\psi_2\psi_3\rangle \langle \psi_2\psi_3| + \lambda_{133} |\psi_3\psi_1\rangle \langle \psi_3\psi_1| + \lambda_{133} |\psi_3\psi_2\rangle \langle \psi_3\psi_2| \\ & + \lambda_{111} |\psi_1\psi_2\psi_3\rangle \langle \psi_1\psi_2\psi_3| + \lambda_{122} |\psi_1\psi_2\psi_3\rangle \langle \psi_1\psi_2\psi_3| + \lambda_{133} |\psi_1\psi_2\psi_3\rangle \langle \psi_1\psi_2\psi_3| \end{aligned} \quad (2)$$

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$$\begin{aligned} & \lambda_{111} |\psi_1\rangle \langle \psi_1| + \lambda_{122} |\psi_2\rangle \langle \psi_2| + \lambda_{133} |\psi_3\rangle \langle \psi_3| \\ & + (\lambda_{111} + \lambda_{122}) |\psi_1\psi_2\rangle \langle \psi_1\psi_2| + (\lambda_{111} + \lambda_{133}) |\psi_1\psi_3\rangle \langle \psi_1\psi_3| + (\lambda_{122} + \lambda_{133}) |\psi_2\psi_3\rangle \langle \psi_2\psi_3| \\ & + (\lambda_{111} + \lambda_{122} + \lambda_{133}) |\psi_1\psi_2\psi_3\rangle \langle \psi_1\psi_2\psi_3| \end{aligned} \quad (3)$$

$$\Gamma(\hat{\lambda}_2) = \sum_{0 \leq N \leq r} \Gamma_N(\hat{\lambda}_2) = \sum_{1 \leq N \leq r} \sum_{1 \leq P \leq N} \binom{N}{P} \Gamma_P^N(\hat{\lambda}_2) = \sum_{2 \leq N \leq 3} \binom{N}{2} \Gamma_2^N(\hat{\lambda}_2) = \hat{\lambda}_2 + 3\Gamma_2^3(\hat{\lambda}_2) \quad (4)$$

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$$\begin{aligned} & \lambda_{21212} |\psi_1\psi_2\rangle \langle \psi_1\psi_2| + \lambda_{21213} |\psi_1\psi_2\rangle \langle \psi_1\psi_3| + \lambda_{21223} |\psi_1\psi_2\rangle \langle \psi_2\psi_3| \\ & + \lambda_{21312} |\psi_1\psi_3\rangle \langle \psi_1\psi_2| + \lambda_{21313} |\psi_1\psi_3\rangle \langle \psi_1\psi_3| + \lambda_{21323} |\psi_1\psi_3\rangle \langle \psi_2\psi_3| \\ & + \lambda_{22312} |\psi_2\psi_3\rangle \langle \psi_1\psi_2| + \lambda_{22313} |\psi_2\psi_3\rangle \langle \psi_1\psi_3| + \lambda_{22323} |\psi_2\psi_3\rangle \langle \psi_2\psi_3| \\ & + (\lambda_{21212} + \lambda_{21313} + \lambda_{22323}) |\psi_1\psi_2\psi_3\rangle \langle \psi_1\psi_2\psi_3| \end{aligned} \quad (5)$$

$$\begin{aligned}
\Gamma\left(\hat{\lambda}_1\right)\Gamma\left(\hat{\lambda}_1\right) &= \left(\hat{\lambda}_1 + 2\Gamma_1^2\left(\hat{\lambda}_1\right) + 3\Gamma_1^3\left(\hat{\lambda}_1\right)\right)\left(\hat{\lambda}_1 + 2\Gamma_1^2\left(\hat{\lambda}_1\right) + 3\Gamma_1^3\left(\hat{\lambda}_1\right)\right) \\
&= \left(\hat{\lambda}_1\right)^2 + 4\Gamma_1^2\left(\hat{\lambda}_1\right)^2 + 9\Gamma_1^3\left(\hat{\lambda}_1\right)^2
\end{aligned} \tag{6}$$

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$$\begin{aligned}
&\lambda_{111}|\psi_1\rangle\langle\psi_1| + \lambda_{122}|\psi_2\rangle\langle\psi_2| + \lambda_{133}|\psi_3\rangle\langle\psi_3| \\
&+ (\lambda_{111} + \lambda_{122})|\psi_1\psi_2\rangle\langle\psi_1\psi_2| + (\lambda_{111} + \lambda_{133})|\psi_1\psi_3\rangle\langle\psi_1\psi_3| + (\lambda_{122} + \lambda_{133})|\psi_2\psi_3\rangle\langle\psi_2\psi_3| \\
&+ (\lambda_{111} + \lambda_{122} + \lambda_{133})|\psi_1\psi_2\psi_3\rangle\langle\psi_1\psi_2\psi_3|
\end{aligned} \tag{7}$$

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$$\begin{aligned}
&\lambda_{111}|\psi_1\rangle\langle\psi_1| + \lambda_{122}|\psi_2\rangle\langle\psi_2| + \lambda_{133}|\psi_3\rangle\langle\psi_3| \\
&+ (\lambda_{111} + \lambda_{122})|\psi_1\psi_2\rangle\langle\psi_1\psi_2| + (\lambda_{111} + \lambda_{133})|\psi_1\psi_3\rangle\langle\psi_1\psi_3| + (\lambda_{122} + \lambda_{133})|\psi_2\psi_3\rangle\langle\psi_2\psi_3| \\
&+ (\lambda_{111} + \lambda_{122} + \lambda_{133})|\psi_1\psi_2\psi_3\rangle\langle\psi_1\psi_2\psi_3|
\end{aligned} \tag{8}$$

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$$\begin{aligned}
&\lambda_{111}^2|\psi_1\rangle\langle\psi_1| + \lambda_{122}^2|\psi_2\rangle\langle\psi_2| + \lambda_{133}^2|\psi_3\rangle\langle\psi_3| \\
&+ (\lambda_{111} + \lambda_{122})^2|\psi_1\psi_2\rangle\langle\psi_1\psi_2| + (\lambda_{111} + \lambda_{133})^2|\psi_1\psi_3\rangle\langle\psi_1\psi_3| + (\lambda_{122} + \lambda_{133})^2|\psi_2\psi_3\rangle\langle\psi_2\psi_3| \\
&+ (\lambda_{111} + \lambda_{122} + \lambda_{133})^2|\psi_1\psi_2\psi_3\rangle\langle\psi_1\psi_2\psi_3|
\end{aligned} \tag{9}$$

$$\begin{aligned}
\Gamma\left(\hat{\lambda}_1\right)\Gamma\left(\hat{\lambda}_2\right) &= \left(\hat{\lambda}_1 + 2\Gamma_1^2\left(\hat{\lambda}_1\right) + 3\Gamma_1^3\left(\hat{\lambda}_1\right)\right)\left(\hat{\lambda}_2 + 3\Gamma_2^3\left(\hat{\lambda}_2\right)\right) \\
&= 2\Gamma_1^2\left(\hat{\lambda}_1\right)\hat{\lambda}_2 + 9\Gamma_1^3\left(\hat{\lambda}_1\right)\Gamma_2^3\left(\hat{\lambda}_2\right)
\end{aligned} \tag{10}$$

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$$\begin{aligned}
& \lambda_{111} |\psi_1\rangle \langle \psi_1| + \lambda_{122} |\psi_2\rangle \langle \psi_2| + \lambda_{133} |\psi_3\rangle \langle \psi_3| \\
& + (\lambda_{111} + \lambda_{122}) |\psi_1\psi_2\rangle \langle \psi_1\psi_2| + (\lambda_{111} + \lambda_{133}) |\psi_1\psi_3\rangle \langle \psi_1\psi_3| + (\lambda_{122} + \lambda_{133}) |\psi_2\psi_3\rangle \langle \psi_2\psi_3| \\
& + (\lambda_{111} + \lambda_{122} + \lambda_{133}) |\psi_1\psi_2\psi_3\rangle \langle \psi_1\psi_2\psi_3|
\end{aligned} \tag{11}$$

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$$\begin{aligned}
& \lambda_{21212} |\psi_1\psi_2\rangle \langle \psi_1\psi_2| + \lambda_{21213} |\psi_1\psi_2\rangle \langle \psi_1\psi_3| + \lambda_{21223} |\psi_1\psi_2\rangle \langle \psi_2\psi_3| \\
& + \lambda_{21312} |\psi_1\psi_3\rangle \langle \psi_1\psi_2| + \lambda_{21313} |\psi_1\psi_3\rangle \langle \psi_1\psi_3| + \lambda_{21323} |\psi_1\psi_3\rangle \langle \psi_2\psi_3| \\
& + \lambda_{22312} |\psi_2\psi_3\rangle \langle \psi_1\psi_2| + \lambda_{22313} |\psi_2\psi_3\rangle \langle \psi_1\psi_3| + \lambda_{22323} |\psi_2\psi_3\rangle \langle \psi_2\psi_3| \\
& + (\lambda_{21212} + \lambda_{21313} + \lambda_{22323}) |\psi_1\psi_2\psi_3\rangle \langle \psi_1\psi_2\psi_3|
\end{aligned} \tag{12}$$

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$$\begin{aligned}
& (\lambda_{111} + \lambda_{122}) \lambda_{21212} |\psi_1\psi_2\rangle \langle \psi_1\psi_2| + (\lambda_{111} + \lambda_{122}) \lambda_{21213} |\psi_1\psi_2\rangle \langle \psi_1\psi_3| + (\lambda_{111} + \lambda_{122}) \lambda_{21223} |\psi_1\psi_2\rangle \langle \psi_2\psi_3| \\
& + (\lambda_{111} + \lambda_{133}) \lambda_{21312} |\psi_1\psi_3\rangle \langle \psi_1\psi_2| + (\lambda_{111} + \lambda_{133}) \lambda_{21313} |\psi_1\psi_3\rangle \langle \psi_1\psi_3| + (\lambda_{111} + \lambda_{133}) \lambda_{21323} |\psi_1\psi_3\rangle \langle \psi_2\psi_3| \\
& + (\lambda_{122} + \lambda_{133}) \lambda_{22312} |\psi_2\psi_3\rangle \langle \psi_1\psi_2| + (\lambda_{122} + \lambda_{133}) \lambda_{22313} |\psi_2\psi_3\rangle \langle \psi_1\psi_3| + (\lambda_{122} + \lambda_{133}) \lambda_{22323} |\psi_2\psi_3\rangle \langle \psi_2\psi_3| \\
& + (\lambda_{111} + \lambda_{122} + \lambda_{133}) (\lambda_{21212} + \lambda_{21313} + \lambda_{22323}) |\psi_1\psi_2\psi_3\rangle \langle \psi_1\psi_2\psi_3|
\end{aligned} \tag{13}$$

$$\begin{aligned}
\Gamma(\hat{\lambda}_2) \Gamma(\hat{\lambda}_1) &= \left(\Gamma(\hat{\lambda}_1) \Gamma(\hat{\lambda}_2) \right)^\dagger = \left(\hat{\lambda}_2 + 3\Gamma_2^3(\hat{\lambda}_2) \right) \left(\hat{\lambda}_1 + 2\Gamma_1^2(\hat{\lambda}_1) + 3\Gamma_1^3(\hat{\lambda}_1) \right) \\
&= 2\hat{\lambda}_2 \Gamma_1^2(\hat{\lambda}_1) + 9\Gamma_2^3(\hat{\lambda}_2) \Gamma_1^3(\hat{\lambda}_1)
\end{aligned} \tag{14}$$

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$$\begin{aligned}
& (\lambda_{111} + \lambda_{122}) \lambda_{21212} |\psi_1\psi_2\rangle \langle \psi_1\psi_2| + (\lambda_{111} + \lambda_{122}) \lambda_{21312} |\psi_1\psi_3\rangle \langle \psi_1\psi_2| + (\lambda_{111} + \lambda_{122}) \lambda_{22312} |\psi_2\psi_3\rangle \langle \psi_1\psi_2| \\
& + (\lambda_{111} + \lambda_{133}) \lambda_{21213} |\psi_1\psi_2\rangle \langle \psi_1\psi_3| + (\lambda_{111} + \lambda_{133}) \lambda_{21313} |\psi_1\psi_3\rangle \langle \psi_1\psi_3| + (\lambda_{111} + \lambda_{133}) \lambda_{22313} |\psi_2\psi_3\rangle \langle \psi_1\psi_3| \\
& + (\lambda_{122} + \lambda_{133}) \lambda_{21223} |\psi_1\psi_2\rangle \langle \psi_2\psi_3| + (\lambda_{122} + \lambda_{133}) \lambda_{21323} |\psi_1\psi_3\rangle \langle \psi_2\psi_3| + (\lambda_{122} + \lambda_{133}) \lambda_{22323} |\psi_2\psi_3\rangle \langle \psi_2\psi_3| \\
& + (\lambda_{111} + \lambda_{122} + \lambda_{133}) (\lambda_{21212} + \lambda_{21313} + \lambda_{22323}) |\psi_1\psi_2\psi_3\rangle \langle \psi_1\psi_2\psi_3|
\end{aligned} \tag{15}$$

$$\begin{aligned}\Gamma\left(\hat{\lambda}_2\right)\Gamma\left(\hat{\lambda}_2\right) &= \left(\hat{\lambda}_2 + 3\Gamma_2^3\left(\hat{\lambda}_2\right)\right)\left(\hat{\lambda}_2 + 3\Gamma_2^3\left(\hat{\lambda}_2\right)\right) \\ &= \left(\hat{\lambda}_2\right)^2 + 9\Gamma_2^3\left(\hat{\lambda}_2\right)^2\end{aligned}\tag{16}$$

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$$\begin{aligned}&\lambda_{21212}|\psi_1\psi_2\rangle\langle\psi_1\psi_2| + \lambda_{21213}|\psi_1\psi_2\rangle\langle\psi_1\psi_3| + \lambda_{21223}|\psi_1\psi_2\rangle\langle\psi_2\psi_3| \\ &+ \lambda_{21312}|\psi_1\psi_3\rangle\langle\psi_1\psi_2| + \lambda_{21313}|\psi_1\psi_3\rangle\langle\psi_1\psi_3| + \lambda_{21323}|\psi_1\psi_3\rangle\langle\psi_2\psi_3| \\ &+ \lambda_{22312}|\psi_2\psi_3\rangle\langle\psi_1\psi_2| + \lambda_{22313}|\psi_2\psi_3\rangle\langle\psi_1\psi_3| + \lambda_{22323}|\psi_2\psi_3\rangle\langle\psi_2\psi_3| \\ &+ (\lambda_{21212} + \lambda_{21313} + \lambda_{22323})|\psi_1\psi_2\psi_3\rangle\langle\psi_1\psi_2\psi_3|\end{aligned}\tag{17}$$

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$$\begin{aligned}&\lambda_{21212}|\psi_1\psi_2\rangle\langle\psi_1\psi_2| + \lambda_{21213}|\psi_1\psi_2\rangle\langle\psi_1\psi_3| + \lambda_{21223}|\psi_1\psi_2\rangle\langle\psi_2\psi_3| \\ &+ \lambda_{21312}|\psi_1\psi_3\rangle\langle\psi_1\psi_2| + \lambda_{21313}|\psi_1\psi_3\rangle\langle\psi_1\psi_3| + \lambda_{21323}|\psi_1\psi_3\rangle\langle\psi_2\psi_3| \\ &+ \lambda_{22312}|\psi_2\psi_3\rangle\langle\psi_1\psi_2| + \lambda_{22313}|\psi_2\psi_3\rangle\langle\psi_1\psi_3| + \lambda_{22323}|\psi_2\psi_3\rangle\langle\psi_2\psi_3| \\ &+ (\lambda_{21212} + \lambda_{21313} + \lambda_{22323})|\psi_1\psi_2\psi_3\rangle\langle\psi_1\psi_2\psi_3|\end{aligned}\tag{18}$$

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$$\begin{aligned}&(\lambda_2)_{1212}^2|\psi_1\psi_2\rangle\langle\psi_1\psi_2| + (\lambda_2)_{1213}^2|\psi_1\psi_2\rangle\langle\psi_1\psi_3| + (\lambda_2)_{1223}^2|\psi_1\psi_2\rangle\langle\psi_2\psi_3| \\ &+ (\lambda_2)_{1312}^2|\psi_1\psi_3\rangle\langle\psi_1\psi_2| + (\lambda_2)_{1313}^2|\psi_1\psi_3\rangle\langle\psi_1\psi_3| + (\lambda_2)_{1323}^2|\psi_1\psi_3\rangle\langle\psi_2\psi_3| \\ &+ (\lambda_2)_{2312}^2|\psi_2\psi_3\rangle\langle\psi_1\psi_2| + (\lambda_2)_{2313}^2|\psi_2\psi_3\rangle\langle\psi_1\psi_3| + (\lambda_2)_{2323}^2|\psi_2\psi_3\rangle\langle\psi_2\psi_3| \\ &+ (\lambda_{21212} + \lambda_{21313} + \lambda_{22323})^2|\psi_1\psi_2\psi_3\rangle\langle\psi_1\psi_2\psi_3|\end{aligned}\tag{19}$$

The squared operator is

$$\begin{aligned}\Gamma\left(\hat{\lambda}_1 + \hat{\lambda}_2\right)\Gamma\left(\hat{\lambda}_1 + \hat{\lambda}_2\right) &= \Gamma\left(\hat{\lambda}_1\right)^2 + \Gamma\left(\hat{\lambda}_1\right)\Gamma\left(\hat{\lambda}_2\right) + \Gamma\left(\hat{\lambda}_2\right)\Gamma\left(\hat{\lambda}_1\right) + \Gamma\left(\hat{\lambda}_2\right)^2 \\ &= \left(\hat{\lambda}_1\right)^2 + 4\Gamma_1^2\left(\hat{\lambda}_1\right)^2 + 9\Gamma_1^3\left(\hat{\lambda}_1\right)^2 + 2\Gamma_1^2\left(\hat{\lambda}_1\right)\hat{\lambda}_2 + 9\Gamma_1^3\left(\hat{\lambda}_1\right)\Gamma_2^3\left(\hat{\lambda}_2\right) + 2\hat{\lambda}_2\Gamma_1^2\left(\hat{\lambda}_1\right) + 9\Gamma_2^3\left(\hat{\lambda}_2\right)\Gamma_1^3\left(\hat{\lambda}_1\right) \\ &+ \left(\hat{\lambda}_2\right)^2 + 9\Gamma_2^3\left(\hat{\lambda}_2\right)^2\end{aligned}\tag{20}$$

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$$\begin{aligned}
& \lambda_{111}^2 |\psi_1\rangle \langle \psi_1| + \lambda_{122}^2 |\psi_2\rangle \langle \psi_2| + \lambda_{133}^2 |\psi_3\rangle \langle \psi_3| \\
& + (\lambda_{111} + \lambda_{122})^2 |\psi_1\psi_2\rangle \langle \psi_1\psi_2| + (\lambda_{111} + \lambda_{133})^2 |\psi_1\psi_3\rangle \langle \psi_1\psi_3| + (\lambda_{122} + \lambda_{133})^2 |\psi_2\psi_3\rangle \langle \psi_2\psi_3| \\
& + (\lambda_{111} + \lambda_{122} + \lambda_{133})^2 |\psi_1\psi_2\psi_3\rangle \langle \psi_1\psi_2\psi_3|
\end{aligned} \tag{21}$$

+

$$\begin{aligned}
& (\lambda_{111} + \lambda_{122}) \lambda_{21212} |\psi_1\psi_2\rangle \langle \psi_1\psi_2| + (\lambda_{111} + \lambda_{122}) \lambda_{21213} |\psi_1\psi_2\rangle \langle \psi_1\psi_3| + (\lambda_{111} + \lambda_{122}) \lambda_{21223} |\psi_1\psi_2\rangle \langle \psi_2\psi_3| \\
& + (\lambda_{111} + \lambda_{133}) \lambda_{21312} |\psi_1\psi_3\rangle \langle \psi_1\psi_2| + (\lambda_{111} + \lambda_{133}) \lambda_{21313} |\psi_1\psi_3\rangle \langle \psi_1\psi_3| + (\lambda_{111} + \lambda_{133}) \lambda_{21323} |\psi_1\psi_3\rangle \langle \psi_2\psi_3| \\
& + (\lambda_{122} + \lambda_{133}) \lambda_{22312} |\psi_2\psi_3\rangle \langle \psi_1\psi_2| + (\lambda_{122} + \lambda_{133}) \lambda_{22313} |\psi_2\psi_3\rangle \langle \psi_1\psi_3| + (\lambda_{122} + \lambda_{133}) \lambda_{22323} |\psi_2\psi_3\rangle \langle \psi_2\psi_3|
\end{aligned} \tag{22}$$

+

$$\begin{aligned}
& (\lambda_{111} + \lambda_{122} + \lambda_{133}) (\lambda_{21212} + \lambda_{21313} + \lambda_{22323}) |\psi_1\psi_2\psi_3\rangle \langle \psi_1\psi_2\psi_3| \\
& (\lambda_{111} + \lambda_{122}) \lambda_{21212} |\psi_1\psi_2\rangle \langle \psi_1\psi_2| + (\lambda_{111} + \lambda_{122}) \lambda_{21312} |\psi_1\psi_3\rangle \langle \psi_1\psi_2| + (\lambda_{111} + \lambda_{122}) \lambda_{22312} |\psi_2\psi_3\rangle \langle \psi_1\psi_2| \\
& + (\lambda_{111} + \lambda_{133}) \lambda_{21213} |\psi_1\psi_2\rangle \langle \psi_1\psi_3| + (\lambda_{111} + \lambda_{133}) \lambda_{21313} |\psi_1\psi_3\rangle \langle \psi_1\psi_3| + (\lambda_{111} + \lambda_{133}) \lambda_{22313} |\psi_2\psi_3\rangle \langle \psi_1\psi_3| \\
& + (\lambda_{122} + \lambda_{133}) \lambda_{21223} |\psi_1\psi_2\rangle \langle \psi_2\psi_3| + (\lambda_{122} + \lambda_{133}) \lambda_{21323} |\psi_1\psi_3\rangle \langle \psi_2\psi_3| + (\lambda_{122} + \lambda_{133}) \lambda_{22323} |\psi_2\psi_3\rangle \langle \psi_2\psi_3| \\
& + (\lambda_{111} + \lambda_{122} + \lambda_{133}) (\lambda_{21212} + \lambda_{21313} + \lambda_{22323}) |\psi_1\psi_2\psi_3\rangle \langle \psi_1\psi_2\psi_3|
\end{aligned} \tag{23}$$

+

$$\begin{aligned}
& (\lambda_2)_{1212}^2 |\psi_1\psi_2\rangle \langle \psi_1\psi_2| + (\lambda_2)_{1213}^2 |\psi_1\psi_2\rangle \langle \psi_1\psi_3| + (\lambda_2)_{1223}^2 |\psi_1\psi_2\rangle \langle \psi_2\psi_3| \\
& + (\lambda_2)_{1312}^2 |\psi_1\psi_3\rangle \langle \psi_1\psi_2| + (\lambda_2)_{1313}^2 |\psi_1\psi_3\rangle \langle \psi_1\psi_3| + (\lambda_2)_{1323}^2 |\psi_1\psi_3\rangle \langle \psi_2\psi_3| \\
& + (\lambda_2)_{2312}^2 |\psi_2\psi_3\rangle \langle \psi_1\psi_2| + (\lambda_2)_{2313}^2 |\psi_2\psi_3\rangle \langle \psi_1\psi_3| + (\lambda_2)_{2323}^2 |\psi_2\psi_3\rangle \langle \psi_2\psi_3| \\
& + (\lambda_{21212} + \lambda_{21313} + \lambda_{22323})^2 |\psi_1\psi_2\psi_3\rangle \langle \psi_1\psi_2\psi_3|
\end{aligned} \tag{24}$$

The difference operator, Δ , is

$$\begin{aligned}
\Delta &= \Gamma(\hat{\lambda}_1 + \hat{\lambda}_2)^2 - \Gamma(\hat{\lambda}_1 + \hat{\lambda}_2) = \Gamma(\hat{\lambda}_1)^2 + \Gamma(\hat{\lambda}_1) \Gamma(\hat{\lambda}_2) + \Gamma(\hat{\lambda}_2) \Gamma(\hat{\lambda}_1) + \Gamma(\hat{\lambda}_2)^2 - \Gamma(\hat{\lambda}_1) - \Gamma(\hat{\lambda}_2) \\
&= (\hat{\lambda}_1)^2 + 4\Gamma_1^2 (\hat{\lambda}_1)^2 + 9\Gamma_1^3 (\hat{\lambda}_1)^2 + 2\Gamma_1^2 (\hat{\lambda}_1) \hat{\lambda}_2 + 9\Gamma_1^3 (\hat{\lambda}_1) \Gamma_2^3 (\hat{\lambda}_2) + 2\hat{\lambda}_2 \Gamma_1^2 (\hat{\lambda}_1) + 9\Gamma_2^3 (\hat{\lambda}_2) \Gamma_1^3 (\hat{\lambda}_1) \\
&+ (\hat{\lambda}_2)^2 + 9\Gamma_2^3 (\hat{\lambda}_2)^2 - \hat{\lambda}_1 - 2\Gamma_1^2 (\hat{\lambda}_1) - 3\Gamma_1^3 (\hat{\lambda}_1) - \hat{\lambda}_2 - 3\Gamma_2^3 (\hat{\lambda}_2)
\end{aligned} \tag{25}$$

$$\Delta = \left(\hat{\lambda}_1\right)^2 - \hat{\lambda}_1 \quad (26)$$

$$+ \left(\hat{\lambda}_2\right)^2 + 4\Gamma_1^2 \left(\hat{\lambda}_1\right)^2 + 2\Gamma_1^2 \left(\hat{\lambda}_1\right) \hat{\lambda}_2 + 2\hat{\lambda}_2\Gamma_1^2 \left(\hat{\lambda}_1\right) - 2\Gamma_1^2 \left(\hat{\lambda}_1\right) - \hat{\lambda}_2 \quad (27)$$

$$+ 9\Gamma_1^3 \left(\hat{\lambda}_1\right) \Gamma_2^3 \left(\hat{\lambda}_2\right) + 9\Gamma_2^3 \left(\hat{\lambda}_2\right) \Gamma_1^3 \left(\hat{\lambda}_1\right) + 9\Gamma_2^3 \left(\hat{\lambda}_2\right)^2 - 3\Gamma_1^3 \left(\hat{\lambda}_1\right) - 3\Gamma_2^3 \left(\hat{\lambda}_2\right) \quad (28)$$

$$\Delta_1 = 0 \Rightarrow \hat{\lambda}_1 = P_{1\nu} \quad (29)$$

$$\Delta_2 = 0 \Rightarrow \quad (30)$$

$$\left(\hat{\lambda}_2\right)^2 + 4P_{2\nu} + 2P_{2\nu}\hat{\lambda}_2 + 2\hat{\lambda}_2P_{2\nu} - 2P_{2\nu} - \hat{\lambda}_2 = \left(\hat{\lambda}_2\right)^2 + 2P_{2\nu} + 2P_{2\nu}\hat{\lambda}_2 + 2\hat{\lambda}_2P_{2\nu} - \hat{\lambda}_2 = 0 \Rightarrow \quad (31)$$

$$P_{2\nu} \left(\hat{\lambda}_2\right)^2 P_{2\nu} + 2P_{2\nu} + 3P_{2\nu}\hat{\lambda}_2P_{2\nu} = 0; P_{2\nu}^\perp \left(\hat{\lambda}_2\right)^2 P_{2\nu}^\perp - P_{2\nu}^\perp \left(\hat{\lambda}_2\right) P_{2\nu}^\perp = 0; \quad (32)$$

$$P_{2\nu}^\perp \left(\hat{\lambda}_2\right)^2 P_{2\nu} + P_{2\nu}^\perp \hat{\lambda}_2 P_{2\nu} = 0 \quad (33)$$

$$\Delta_3 = 0 \Rightarrow \quad (34)$$

$$3P_{3\nu}\Gamma_2^3 \left(\hat{\lambda}_2\right) + 3\Gamma_2^3 \left(\hat{\lambda}_2\right) P_{3\nu} + 3\Gamma_2^3 \left(\hat{\lambda}_2\right)^2 - \Gamma_2^3 \left(\hat{\lambda}_2\right) = P_{3\nu} \Rightarrow \quad (35)$$

$$5P_{3\nu}\Gamma_2^3 \left(\hat{\lambda}_2\right) P_{3\nu} + 3P_{3\nu}\Gamma_2^3 \left(\hat{\lambda}_2\right)^2 P_{3\nu} = P_{3\nu} \quad (36)$$

$$2P_{3\nu}\Gamma_2^3 \left(\hat{\lambda}_2\right) P_{3\nu}^\perp + 3P_{3\nu}\Gamma_2^3 \left(\hat{\lambda}_2\right)^2 P_{3\nu}^\perp = 0 \quad (37)$$

$$3P_{3\nu}^\perp\Gamma_2^3 \left(\hat{\lambda}_2\right)^2 P_{3\nu}^\perp - P_{3\nu}^\perp\Gamma_2^3 \left(\hat{\lambda}_2\right) P_{3\nu}^\perp = 0 \quad (38)$$

r=4*****

$$\Gamma \left(\hat{\lambda}_1\right) \quad (39)$$

||

$$\lambda_{21412} |\psi_1 \psi_4\rangle \langle\psi_1 \psi_2| + \lambda_{21413} |\psi_1 \psi_4\rangle \langle\psi_1 \psi_3| + \lambda_{21414} |\psi_1 \psi_4\rangle \langle\psi_1 \psi_4| + \lambda_{21423} |\psi_1 \psi_4\rangle \langle\psi_2 \psi_3| + \lambda_{21424} |\psi_1 \psi_4\rangle \langle\psi_2 \psi_4| + \dots$$
(45)

$$\lambda_{22312} |\psi_2 \psi_3\rangle \langle \psi_1 \psi_2| + \lambda_{22313} |\psi_2 \psi_3\rangle \langle \psi_1 \psi_3| + \lambda_{22314} |\psi_2 \psi_3\rangle \langle \psi_1 \psi_4| + \lambda_{22323} |\psi_2 \psi_3\rangle \langle \psi_2 \psi_3| + \lambda_{22324} |\psi_2 \psi_3\rangle \langle \psi_2 \psi_4| + \dots$$

$$\lambda_{22412} |\psi_2 \psi_4\rangle \langle \psi_1 \psi_2| + \lambda_{22413} |\psi_2 \psi_4\rangle \langle \psi_1 \psi_3| + \lambda_{22414} |\psi_2 \psi_4\rangle \langle \psi_1 \psi_4| + \lambda_{22423} |\psi_2 \psi_4\rangle \langle \psi_2 \psi_3| + \lambda_{22424} |\psi_2 \psi_4\rangle \langle \psi_2 \psi_4| + \lambda_{22432} |\psi_2 \psi_4\rangle \langle \psi_2 \psi_3| + \lambda_{22433} |\psi_2 \psi_4\rangle \langle \psi_2 \psi_4| + \lambda_{22434} |\psi_2 \psi_4\rangle \langle \psi_2 \psi_4| + \lambda_{22442} |\psi_2 \psi_4\rangle \langle \psi_2 \psi_4| + \lambda_{22443} |\psi_2 \psi_4\rangle \langle \psi_2 \psi_4| + \lambda_{22444} |\psi_2 \psi_4\rangle \langle \psi_2 \psi_4|$$

$$\lambda_{23412} |\psi_3\psi_4\rangle \langle \psi_1\psi_2| + \lambda_{23413} |\psi_3\psi_4\rangle \langle \psi_1\psi_3| + \lambda_{23414} |\psi_3\psi_4\rangle \langle \psi_1\psi_4| + \lambda_{23423} |\psi_3\psi_4\rangle \langle \psi_2\psi_3| + \lambda_{23424} |\psi_3\psi_4\rangle \langle \psi_2\psi_4| + \lambda_{23432} |\psi_3\psi_4\rangle \langle \psi_3\psi_2| + \lambda_{23434} |\psi_3\psi_4\rangle \langle \psi_3\psi_4| + \lambda_{23442} |\psi_3\psi_4\rangle \langle \psi_4\psi_2| + \lambda_{23443} |\psi_3\psi_4\rangle \langle \psi_4\psi_3| + \lambda_{23444} |\psi_3\psi_4\rangle \langle \psi_4\psi_4|$$

$$+ \tag{49}$$

$$\lambda_{21212} |\psi_1 \psi_2 \psi_3\rangle \langle \psi_1 \psi_2 \psi_3| + \lambda_{21212} |\psi_1 \psi_2 \psi_4\rangle \langle \psi_1 \psi_2 \psi_4| + \lambda_{21213} |\psi_1 \psi_2 \psi_4\rangle \langle \psi_1 \psi_3 \psi_4| + \lambda_{21214} |\psi_1 \psi_2 \psi_3\rangle \langle \psi_1 \psi_4 \psi_3| \quad (50)$$

$$+ \lambda_{21223} |\psi_1 \psi_2 \psi_4\rangle \langle \psi_2 \psi_3 \psi_4| + \lambda_{21224} |\psi_1 \psi_2 \psi_3\rangle \langle \psi_2 \psi_4 \psi_3| \quad (51)$$

$$\lambda_{21312} |\psi_1 \psi_3 \psi_4\rangle \langle \psi_1 \psi_2 \psi_4| + \lambda_{21313} |\psi_1 \psi_3 \psi_2\rangle \langle \psi_1 \psi_3 \psi_2| + \lambda_{21313} |\psi_1 \psi_3 \psi_4\rangle \langle \psi_1 \psi_3 \psi_4| + \lambda_{21314} |\psi_1 \psi_3 \psi_2\rangle \langle \psi_1 \psi_4 \psi_2| \quad (52)$$

$$+ \lambda_{21323} |\psi_1 \psi_3 \psi_4\rangle \langle \psi_2 \psi_3 \psi_4| + \lambda_{21334} |\psi_1 \psi_3 \psi_2\rangle \langle \psi_3 \psi_4 \psi_2| \quad (53)$$

$$\lambda_{21412} |\psi_1 \psi_4 \psi_3\rangle \langle \psi_1 \psi_2 \psi_3| + \lambda_{21413} |\psi_1 \psi_4 \psi_2\rangle \langle \psi_1 \psi_3 \psi_2| + \lambda_{21414} |\psi_1 \psi_4 \psi_3\rangle \langle \psi_1 \psi_4 \psi_3| + \lambda_{21414} |\psi_1 \psi_4 \psi_2\rangle \langle \psi_1 \psi_4 \psi_2| \quad (54)$$

$$+ \lambda_{21424} |\psi_1 \psi_4 \psi_3\rangle \langle \psi_2 \psi_4 \psi_3| + \lambda_{21434} |\psi_1 \psi_4 \psi_2\rangle \langle \psi_3 \psi_4 \psi_2| \quad (55)$$

$$\lambda_{22312} |\psi_2 \psi_3 \psi_4\rangle \langle \psi_1 \psi_2 \psi_4| + \lambda_{22313} |\psi_2 \psi_3 \psi_4\rangle \langle \psi_1 \psi_3 \psi_4| + \lambda_{22323} |\psi_2 \psi_3 \psi_1\rangle \langle \psi_2 \psi_3 \psi_1| + \lambda_{22323} |\psi_2 \psi_3 \psi_4\rangle \langle \psi_2 \psi_3 \psi_4| \quad (56)$$

$$+ \lambda_{22324} |\psi_2 \psi_3 \psi_1\rangle \langle \psi_2 \psi_4 \psi_1| + \lambda_{22334} |\psi_2 \psi_3 \psi_1\rangle \langle \psi_3 \psi_4 \psi_1| \quad (57)$$

$$\lambda_{22412} |\psi_2 \psi_4 \psi_3\rangle \langle \psi_1 \psi_2 \psi_3| + \lambda_{22414} |\psi_2 \psi_4 \psi_3\rangle \langle \psi_1 \psi_4 \psi_3| + \lambda_{22423} |\psi_2 \psi_4 \psi_1\rangle \langle \psi_2 \psi_3 \psi_1| + \lambda_{22424} |\psi_2 \psi_4 \psi_3\rangle \langle \psi_2 \psi_4 \psi_3| \quad (58)$$

$$+ \lambda_{22424} |\psi_2 \psi_4 \psi_1\rangle \langle \psi_2 \psi_4 \psi_1| + \lambda_{22434} |\psi_2 \psi_4 \psi_1\rangle \langle \psi_3 \psi_4 \psi_1| \quad (59)$$

$$\lambda_{23412} |\psi_3 \psi_4 \psi_1\rangle \langle \psi_1 \psi_2 \psi_1| + \lambda_{23413} |\psi_3 \psi_4 \psi_2\rangle \langle \psi_1 \psi_3 \psi_2| + \lambda_{23423} |\psi_3 \psi_4 \psi_1\rangle \langle \psi_2 \psi_3 \psi_1| + \lambda_{23424} |\psi_3 \psi_4 \psi_1\rangle \langle \psi_2 \psi_4 \psi_1| \quad (60)$$

$$+ \lambda_{22434} |\psi_3 \psi_4 \psi_1\rangle \langle \psi_3 \psi_4 \psi_1| + \lambda_{22434} |\psi_3 \psi_4 \psi_2\rangle \langle \psi_3 \psi_4 \psi_2| \quad (61)$$

$$\parallel \quad (62)$$

$$(\lambda_{21212} + \lambda_{21313} + \lambda_{22323}) |\psi_1 \psi_2 \psi_3\rangle \langle \psi_1 \psi_2 \psi_3| + (\lambda_{21212} + \lambda_{21414} + \lambda_{22424}) |\psi_1 \psi_2 \psi_4\rangle \langle \psi_1 \psi_2 \psi_4| + \quad (63)$$

$$(\lambda_{21313} + \lambda_{21414} + \lambda_{22434}) |\psi_1 \psi_3 \psi_4\rangle \langle \psi_1 \psi_3 \psi_4| + (\lambda_{22323} + \lambda_{22424} + \lambda_{22434}) |\psi_2 \psi_3 \psi_4\rangle \langle \psi_2 \psi_3 \psi_4| + \quad (64)$$

$$(\lambda_{21213} + \lambda_{22434}) |\psi_1 \psi_2 \psi_4\rangle \langle \psi_1 \psi_3 \psi_4| + (\lambda_{22334} - \lambda_{21214}) |\psi_1 \psi_2 \psi_3\rangle \langle \psi_1 \psi_3 \psi_4| + \quad (65)$$

$$(\lambda_{21223} - \lambda_{21434}) |\psi_1 \psi_2 \psi_4\rangle \langle \psi_2 \psi_3 \psi_4| + (\lambda_{21224} - \lambda_{21334}) |\psi_1 \psi_2 \psi_3\rangle \langle \psi_2 \psi_3 \psi_4| + \quad (66)$$

$$(\lambda_{21312} + \lambda_{23424}) |\psi_1 \psi_3 \psi_4\rangle \langle \psi_1 \psi_2 \psi_4| + (\lambda_{21314} + \lambda_{22324}) |\psi_1 \psi_2 \psi_3\rangle \langle \psi_1 \psi_2 \psi_4| + \quad (67)$$

$$(\lambda_{21323} + \lambda_{21424}) |\psi_1 \psi_3 \psi_4\rangle \langle \psi_2 \psi_3 \psi_4| + (\lambda_{23423} - \lambda_{21412}) |\psi_1 \psi_3 \psi_4\rangle \langle \psi_1 \psi_2 \psi_3| + \quad (68)$$

$$(\lambda_{22312} - \lambda_{23414}) |\psi_2 \psi_3 \psi_4\rangle \langle \psi_1 \psi_2 \psi_4| + (\lambda_{22414} - \lambda_{22313}) |\psi_2 \psi_3 \psi_4\rangle \langle \psi_1 \psi_3 \psi_4| + \quad (69)$$

$$(\lambda_{22412} + \lambda_{23413}) |\psi_2 \psi_3 \psi_4\rangle \langle \psi_1 \psi_2 \psi_3| + (\lambda_{22423} + \lambda_{21413}) |\psi_1 \psi_2 \psi_4\rangle \langle \psi_1 \psi_2 \psi_3| \quad (70)$$

$$+ \quad (71)$$

$$(\lambda_{21212} + \lambda_{21313} + \lambda_{21414} + \lambda_{22323} + \lambda_{22424} + \lambda_{23434}) |\psi_1 \psi_2 \psi_3 \psi_4\rangle \langle \psi_1 \psi_2 \psi_3 \psi_4| \quad (72)$$